

**Developing Decision Tree Models for Prediction in Kidney
Post-Transplantation Outcomes**

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Abstract

The development of building intuitive machine learning tools in healthcare has become more prevalent. Successful models hold significant promise for various applications in the medical field. This paper focuses on developing decision tree machine learning models to predict deceased donor kidney post-transplantation outcomes of a recipient contracting cancer or experiencing graft failure in kidney recipients. The decision tree algorithm will be instrumental to the kidney transplantation field in a variety of ways, from pre-transplant evaluation of assessing suitability of patients for transplantation by predicting outcomes, to providing insight on factors that influence deceased donor transplant outcomes. The research first involves investigating features to be selected for the model. The measures for this included the current kidney allocation system and matching policies, previous studies involving patient and surgical opinion, and an ongoing study to gain more insight into a transplant surgeon's decision making process. Then, the decision tree models for predicting cancer and graft failure were evaluated. Both models demonstrated high precision and strength in predicting 'no cancer' and 'no graft failure,' but did not perform well on identifying 'cancer' and 'graft failure' cases. The discrepancy was due to a class imbalance in the dataset. Future research will involve preparing a more relevant and balanced dataset to reevaluate the models, continue discovering important factors in the decision making process of organ transplantation, and creating machine learning tools for other types of organ transplantation.

Background and Introduction

End-stage Renal Disease and Treatment Options

End-stage Renal Disease (ESRD), also commonly referred to as kidney failure, is a significant public health challenge in the United States. This condition occurs when the gradual loss of kidney function reaches an advanced state and a person's kidneys cannot fulfill the body's needs any longer [1]. Under normal conditions, kidneys will filter waste and excess fluids from the blood, which are then excreted in your urine. However, with ESRD, kidney failure results in losing filtering ability which leads to dangerous levels of fluid accumulation. Treatments for ESRD include peritoneal dialysis, which involves inserting a permanent catheter into the abdominal lining for a dialysis solution that is regularly replaced and drained every few hours [2]. Although dialysis is life-sustaining, this treatment option often falls short in improving a patient's quality of life by putting limitations on a patient's schedule and mobility. Thus, many patients have sought kidney transplantation and have opted into the transplant waiting list, with an annual record of over 25,000 kidney transplant operations in 2022 [3].

Organ Allocation Opinion Measures

The United Network for Organ Sharing (UNOS) kidney allocation system, which oversees the Organ Procurement and Transplantation Network (OPTN), aims to provide an approach that addresses the medical complexities of transplantation and optimizes the use of available organs. The desired outcomes encompass maximizing patient survival, matching quality kidneys with specific patients' health states that ensure prolonged and healthy living, and offering the best possible longevity and overall life benefits. The allocation system uses a computerized matching system, UNet, to generate a ranked list of candidates for a donor. The

ranking is generated by point-based merit, with certain medical and logistical factors, implemented in OPTN policies, that give patients medical priority [4].

Maximum differences scaling (MDS) analysis is a survey-based experimental design obtaining preference through importance scores for multiple factors [5]. In an MDS study in medical application, participants are presented with several factors and subsequently asked to identify the factor they deem most important for a given clinical consideration (Figure 1).

Considering only these factors, which do you feel is Most Important in deciding whether or not to accept the kidney?

Factor	Select most important
How closely matched I am with the donor	☐
The age of the donor	☐
How much the donor weighed	☐
How long I would have to wait for another kidney	☒

Figure 1: A potential patient question included in a maximum differences scaling survey.

A discrete choice experiment (DCE) is another survey-based design where preferences are obtained through a series of hypothetical scenarios [6]. Each hypothetical scenario consists of a set of alternatives, which have multiple attribute levels (Figure 2).

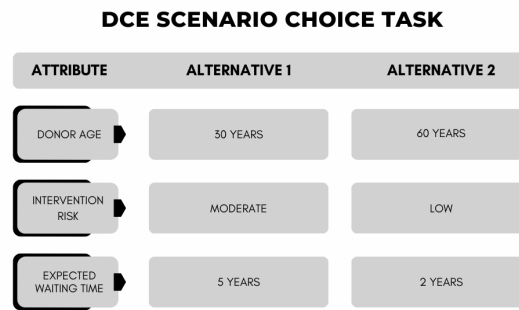


Figure 2: A DCE scenario framework consisting of two alternatives and three attribute levels.

Machine Learning for Kidney Transplantation

Marginal donors, organs with unfavorable characteristics but still viable for transplantation, are often discarded despite having potential to offer survival benefits [7]. By understanding the health status of patients and defining a measure that determines suitability to a donor, the limited supply of kidney donors can be used more effectively.

There have been intuitive online machine learning tools based on national registry data developed to aid medical decisions. Using these models help us visualize and have a better understanding of the interaction between donor quality and candidate condition, and could increase access to kidney transplantation while maximizing utilization of all viable donors.

Specifically, deceased donor kidney transplantation (DDKT) has become increasingly common among transplant candidates [8]. However, unlike living donor kidney transplants which function immediately, patients who follow through with DDKT operations may have to go back on temporary dialysis due to delayed function. More concerning, recipients may experience life threatening post-transplant complications such as graft failure or cancer [9].

In this project, I investigated multiple resources to identify key factors and priorities taken into consideration when deciding to accept a deceased donor. Then, I implemented a

decision tree (DT) algorithm for predicting two critical post DDKT outcomes utilizing data provided by UNOS: contracting cancer and experiencing graft failure. These clinical outcomes are crucial to consider before performing DDKT. These models were assessed using various performance metrics and figures to explore the potential medical applications of these machine learning algorithms.

Materials and Methods

For this study, I utilized a hand-prepared version of UNOS' national registry data regarding all DDKT acceptances from January to March of 2019 (N=3,084). This dataset was utilized to develop machine learning models in predicting post-transplant cancer occurrences and graft failure.

Feature selection was conducted through integrating insights on three patient surgeon references: patient and surgeon responses from an MDS analysis [5], findings from a patient DCE on intervention research [10], and our ongoing research on developing a kidney transplant surgeon DCE. Current UNOS matching and allocation policies were also investigated. These references helped determine critical factors and attributes of both donors and recipients to take into account when receiving a DDKT offer.

Allocation System and Matching Policies

The Kidney Donor Profile Index (KDPI) score, commonly interpreted as a measure of donor quality, ranges from 0-100% and is derived from 10 donor factors (refer to Formula 1, Tables 6 and 7). A low KDPI score indicates a donor of high quality, reflecting the relative

lifespan of the kidney function compared to other donors. The KDPI score emerged as a significant factor in donor evaluations [11].

Calculated panel reactive antibody (cPRA) refers to a percentage that is interpreted as a recipient's sensitization to organ donors, how prone they are to organ rejection. It was also learned that people who have a cPRA higher than 98% are automatically prioritized nationally due to not being able to find many compatible donors [4]. Given that cPRA may be measured and interpreted as the amount of donors a recipient is incompatible with, I concluded it is an important factor in accepting a donor offer.

Patient and Surgeon Perspectives

In the MDS study, participants were asked to consider a certain number of factors and select which factor they believe was most important in accepting a DDKT offer. Additionally, correlation and association analyses were performed between five key patient characteristics and their reported importance scores [5]. The study observed an inverse relationship for association between a patient's time on waiting list and the relative importance they would place on the donor's kidney function and quality.

Among the top nine priorities identified by patients and surgeons, six were shared priorities. This resulted in underscoring the importance of the overall kidney quality, how long the kidney is expected to last, how difficult it is to be matched to a kidney, the function of kidney at time of death, the general health of the donor, and the expected wait time for another donor if a patient were to pass on the offer.

I also looked at a DCE study that focused on intervention organ research and organ acceptance at various risk levels. The study revealed that older participants and those who had

spent less time on dialysis or the waitlist were more inclined to accept intervention organs and participate in organ research.

Lastly, I investigated our current research on developing a DCE survey for kidney transplant surgeons, aiming to discern significant deceased donor characteristics that transplant surgeons consider when faced with a marginal offer. The conducted preliminary interviews to develop the survey with four kidney transplant surgeons suggest additional factors deemed important by surgeons that are not covered in KDPI calculations. This consists of acute kidney injury (AKI), renal flow rate, biopsy results, urine output, human leukocyte antigen (HLA) mismatch level, and cold ischemic time.

Results

Factor	Importance
KDPI (%)	Overall quality of donor kidney
Serum Creatinine Level (mg/dL)	Overall function of kidney
Body Mass Index (BMI)	General health of donor
cPRA (%)	How difficult it is to be matched to a kidney
Time on waiting list (days)	Time on waiting list
Recipient's age (years)	Recipient's age
HLA mismatch level	How closely matched to donor
Cold Ischemic Time (hours)	Overall function of kidney

Table 1: List of the 8 selected features and the importance to include in DT model

A collective total of eight features, shown in Table 1, were derived from the investigation on important factors and were utilized to construct the DT models. This consisted of generalizing results found from previous studies and our current research on developing a DCE. I built two distinct DT models, one which classifies whether or not patients would develop cancer post-transplantation, and a second for predicting if a recipient would experience graft failure post-transplantation.

For comprehensive visualization, DT hierarchies with a test-run maximum depth of 4 were plotted. After the models were constructed, classification reports and confusion matrix analyses were conducted to assess the models' performance.

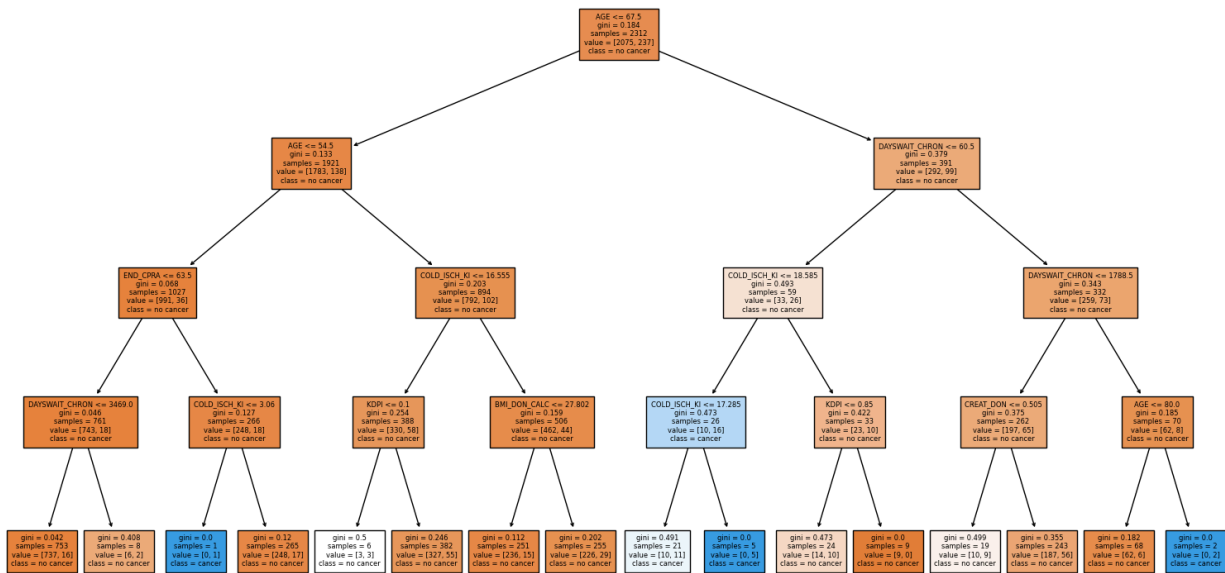


Figure 3: Decision tree hierarchy of maximum depth 4 for predicting cancer post-transplant

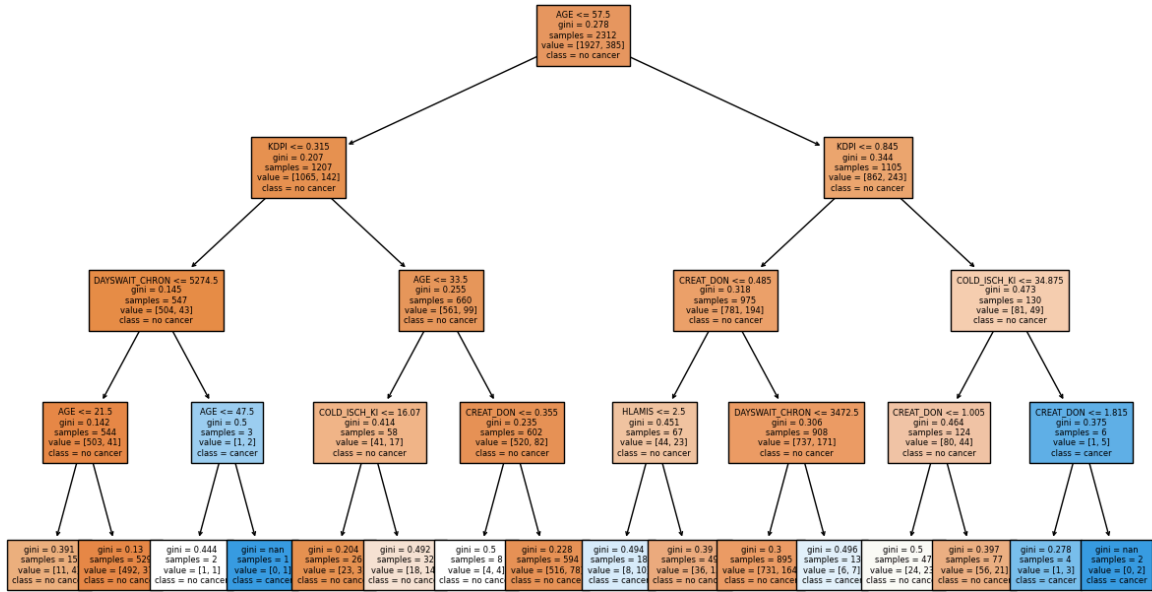


Figure 4: Decision tree hierarchy of maximum depth 4 for predicting graft failure post-transplant

	Precision	Recall
No Cancer	.92	.89
Cancer	.20	.27

Table 2: Precision and recall scores for DT model predicting cancer post-transplant

	Precision	Recall
No Graft Failure	.85	.86
Graft Failure	.18	.17

Table 3: Precision and recall scores for DT model predicting graft failure post-transplant

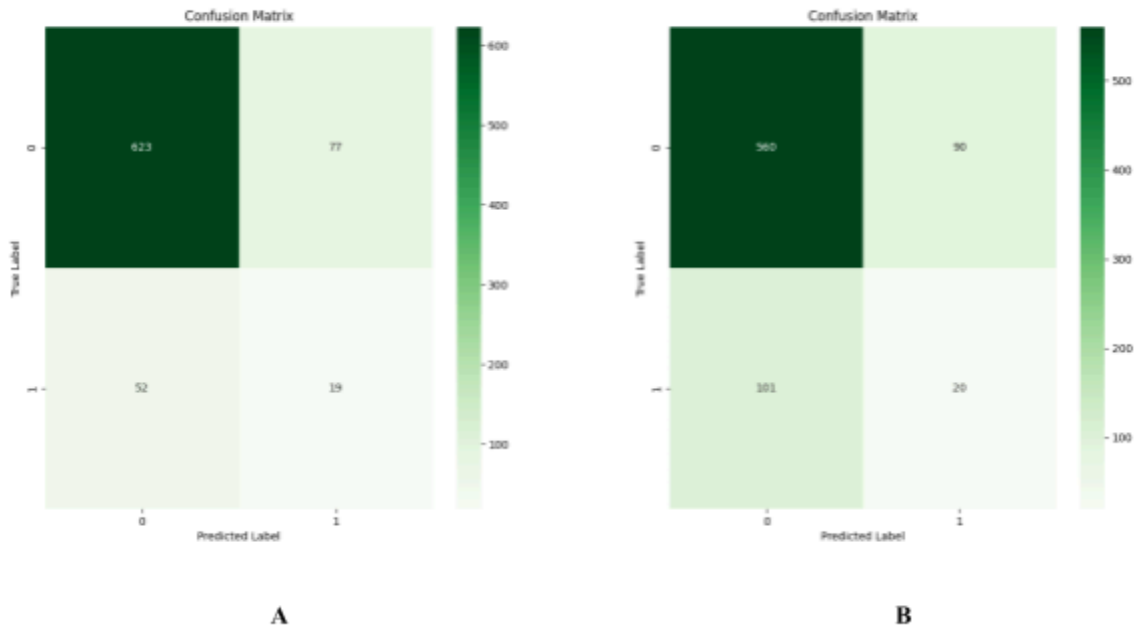


Figure 5: Confusion matrices for DT models. A) Predicting cancer post-transplant. B) Predicting graft failure post-transplant.

Because of the class distributions (Figure 6), I resampled by over-sampling, reran the algorithm, and evaluated its classification report (Table 4, 5) and confusion matrices (Figure 7).

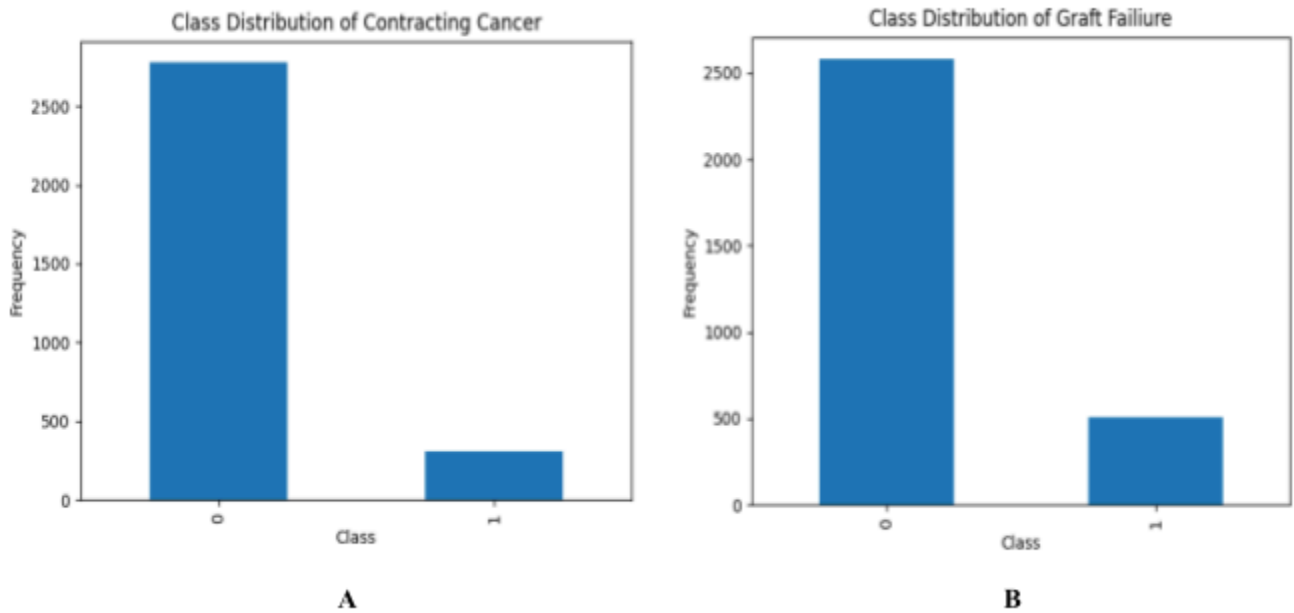


Figure 6: Class distribution for DT models. A) Predicting cancer post-transplant. B) Predicting graft failure post-transplant.

	Precision	Recall
No Cancer	.96	.59
Cancer	.16	.75

Table 4: Precision and recall scores for DT model predicting cancer post-transplant using a resampled dataset.

	Precision	Recall
No Graft Failure	.91	.52
Graft Failure	.22	.72

Table 5: Precision and recall scores for DT model predicting graft failure post-transplant using a resampled dataset.

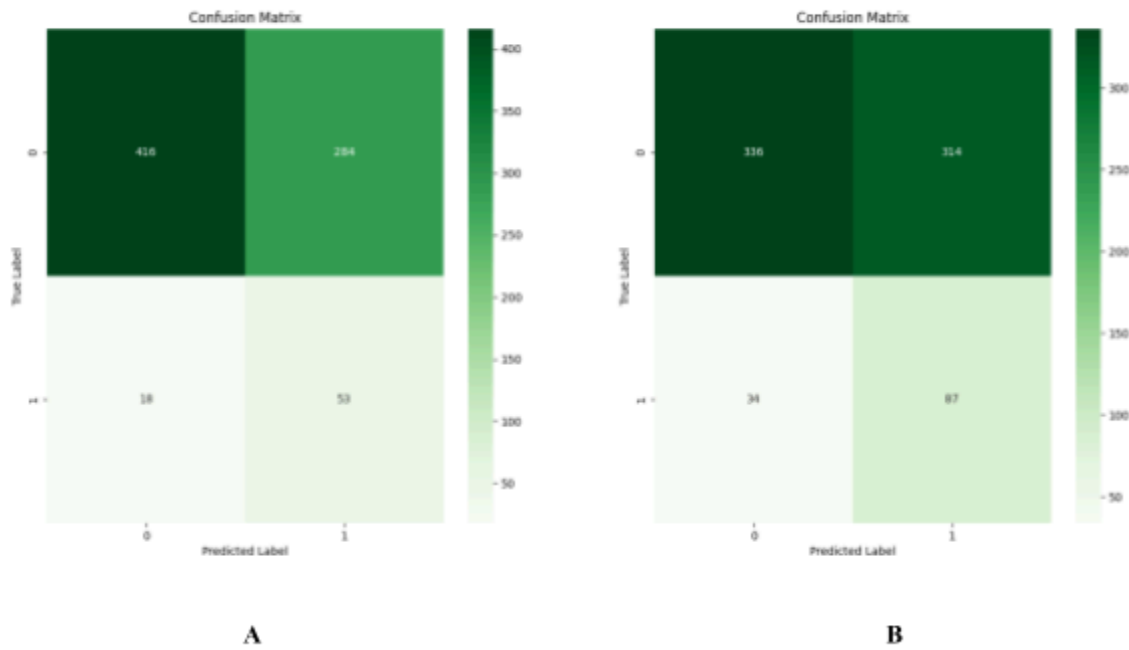


Figure 7: Confusion matrices for DT models using a resampled dataset. A) Predicting cancer post-transplant. B) Predicting graft failure post-transplant.

Discussion

Decision Tree Models

The cancer DT model performed with a precision of .92 for ‘no cancer’ predictions and .20 for ‘cancer’ predictions. This implies that while the model is highly reliable in correctly identifying non-cancer cases, its ability to accurately predict cancer cases is considerably lower. The model’s recall was .89 for ‘no cancer’ and .27 for ‘cancer’, indicating it successfully identified 89% of non-cancer cases but only 27% of actual cancer cases (Table 2). Although the model shows strength in predicting and identifying non-cancer cases, these results are misleading to the overall performance of the model. In the training set, 52 cancer instances were classified as no cancer, while only 19 cancer instances were predicted accurately (Figure 5A).

The graft failure DT model's precision was .85 for 'no failure' and .18 for 'failure'. Precision reflects the model's accuracy in predicting true positive results, and the graft failure DT is more reliable in predicting 'no graft failure' cases compared to actual failures. Its recall rates were .86 for 'no failure' and .17 for 'failure', signifying its capacity to correctly identify 86% of non-failure cases, but only 17% of actual failures (Table 3). This is concerning, as the model incorrectly classified 101 instances of graft failure as 'no failure', a significant issue in the context of life-saving medical decisions (Figure 5B).

For significant medical decisions that determine the life or death of a recipient and a success or failure of a DDKT, it's important that the model performs with a high recall value. Unfortunately, this was not achieved in our models. Shown in Figure 6, the class distributions for cancer and graft failure were imbalanced with a predominance of 'no cancer' and 'no graft failure' instances. This skewed the DT models' ability to accurately predict cancer and graft failure events and suggests why the model performed poorly in correctly identifying these occurrences in the dataset.

To counter the class imbalance and considering the trade-offs between precision and recall, I prioritized recall, even so at the expense of precision, because in medical application, it is more crucial to avoid missing actual cases of cancer and graft failure. Consequently, two additional DT algorithms were implemented using an over-sampling technique. The recall for these models improved to .75 and .72 for cancer and graft failure cases respectively (Tables 4, 5). However, the models exhibited a tendency to misclassify 'no cancer' cases as 'cancer' (Figure 7), and thus, suboptimal for maximizing kidney utilization.

It's also worth noting that the recipient's age consistently emerged as the root node and primary split factor in all DT models that were built (Figure 3, 4). More notably, the paths of the

older age led to more leaf nodes associated with cancer and graft failure occurrences in comparison to younger recipients. Given a relatively normal distribution of age (Figure 8) across the dataset, the validity and intuitivity of this trend should be further investigated.

Limitations

There were several limitations to this research. In the feature selection process, relying on patient and surgeon input from the MDS and DCE studies may not have offered a full review and perspective on all important attributes. This raises questions about the generalizability of the findings and their applicability across all clinical settings.

Regarding the current allocation system and its policies, in developing our current DCE, many transplant surgeons indicated the KDPI metric has many values that are weighted improperly and misrepresented within the equation. This may influence the accuracy of this research and the effectiveness of utilizing KDPI as a main measure in clinical scenarios.

The dataset used had many shortcomings that led to inconclusive results in evaluating the true potential of machine learning tools in kidney transplantation research. Because the dataset was imbalanced, the DT models created were not able to perform well on classifying and identifying ‘cancer’ and ‘graft failure’ cases due to such a small number of instances to train on compared to ‘no cancer’ and ‘no graft failure’ cases.

Furthermore, the dataset did include the many attributes that were found important and as priority from patient and surgeon opinion. This included factors such as urine output which is a measure in the overall function and health of a kidney donor and considered an important factor by transplant surgeons from conducted interviews, as well as AKI, renal flow rate, and biopsy

results. I wasn't able to implement all the factors I wanted into the algorithm, which could have affected the models' performance of the model and intuitiveness.

Future Research and Applications

In summary, while the model shows good performance in identifying 'no cancer' cases, it struggles significantly with accurately identifying 'cancer' cases, which is crucial in a medical context. The next step in this research should involve hand-preparing and reprocessing a new, relevant, and balanced dataset that better represents the diversity of post-transplant outcomes. Although this process is time consuming, it is crucial for developing more accurate models. By addressing these dataset limitations, future iterations of the models could offer more reliable and clinically relevant predictions.

I also plan to continue developing the transplant surgeon DCE to better gain a deeper understanding of the priorities of transplant surgeons when evaluating a DDKT offer. By better aligning the model with the priorities and practical considerations of transplant surgeons, we can enhance the models' relevance and applicability in real-world clinical settings.

Lastly, given the increasing trend in organ transplantation and the limited supply of organs, in the future, I look to implement machine learning models for other organ transplantations and their associated risks. Developing models that can accurately predict post-transplantation outcomes for various organs not only would help in understanding the nuances and unique risk factors associated with different types of organ transplants, but would also significantly enhance our ability to make informed medical decisions, optimize organ utilization, and save more lives.

Acknowledgements

I would like to thank Dr. Michael Fu (The University of Maryland, College Park) for granting me the opportunity to collaborate with his students over the summer and for his advisory with the research process. I would also like to thank Mr. Xingyu Ren for his mentorship throughout this project, Mr. Zhenyu Yue for familiarizing me with the kidney transplantation field, and Mr. Sanketh Andhavarapu for his valuable insights on the current DCE study. Lastly, I want to thank Ms. Angelique Bosse for her guidance and my peers for providing invaluable comments.

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Appendix

Figure 8: Distribution of Recipient’s Age in the dataset

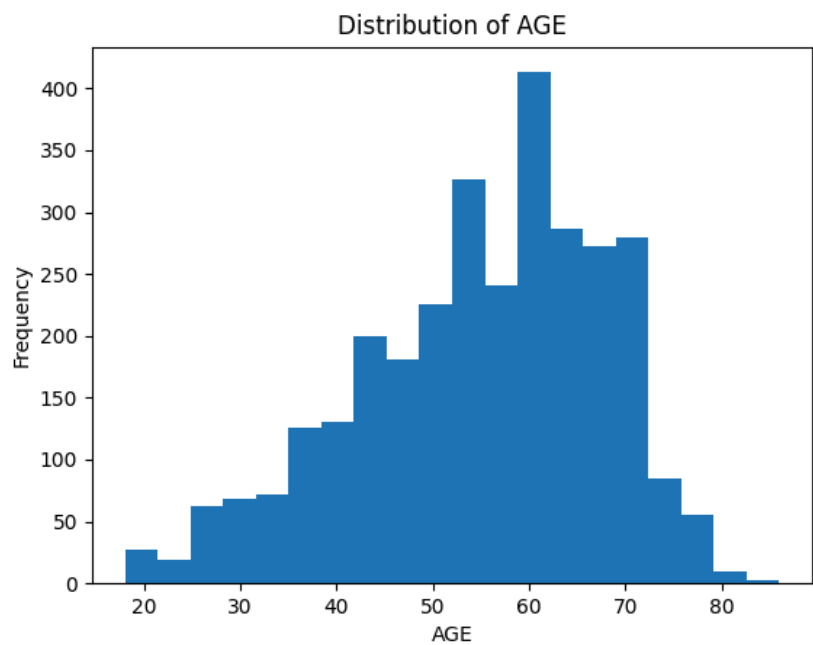


Table 6: List of KDRI Score Components

Donor Characteristic	Applies to Which Donors
Age (integer)	All
	Age < 18
	Age > 50
Height (cm)	All
Weight (kg)	Weight < 80 kg
Ethnicity	African American
History of Hypertension	Hypertensive
History of Diabetes	Diabetic
Cause of Death	Cerebrovascular Accident
Serum Creatinine (mg/dL)	All

	Creatinine > 1.5
Hepatitis C Virus Status (HCV)	HCV Positive
Donation of Circulatory Death (DCD)	DCD

Formula 1: Formula X utilized to calculate KDRI

$X = \sum \text{KDRI Score Components}$

$X = 0.0128 \times (\text{age} - 40) - 0.0194 \times (\text{age} - 18) \times I(\text{age} < 18) + 0.0107 \times (\text{age} - 50) \times I(\text{age} > 50)$
 $\text{height} - 170 \text{ weight} - 80 - 0.0464 \times - 0.0199 \times \times I(\text{weight} < 80 \text{ kg}) 10.5 + 0.1790 \times I(\text{African}$
 $\text{American ethnicity}) + 0.1260 \times I(\text{History of hypertension}) + 0.1300 \times I(\text{History of diabetes}) +$
 $0.0881 \times I(\text{COD} = \text{CVA}) + 0.2200 \times (\text{Creatinine} - 1) - 0.2090 \times (\text{Creatinine} - 1) \times I(\text{Creatinine}$
 $> 1.5 \text{ mg/dL}) + 0.2400 \times I(\text{HCV positive}) + 0.1330 \times I(\text{DCD})$

$$\text{KDRI}_{\text{rao}} = e^X$$

$$\text{KDRI} = \text{KRI}_{\text{rao}} / (\text{donor median KRI}_{\text{rao}})$$

Table 7: 2023 OPTN KDRI-to-KDPI Mapping Table

Source: Organ Procurement and Transplantation Network, 2023.

https://optn.transplant.hrsa.gov/media/wmnxxzu/kdpi_mapping_table.pdf

KDRI>	KDRI<=	KDPI
0.0000000000000000	0.45068731085587	0%
0.45068731085587	0.53693152638850	1%
0.53693152638850	0.56012914599981	2%
0.56012914599981	0.57594408135238	3%
0.57594408135238	0.59019546284842	4%
0.59019546284842	0.60390915137906	5%
0.60390915137906	0.61387036565312	6%
0.61387036565312	0.62571423608426	7%

0.62571423608426	0.63467545579118	8%
0.63467545579118	0.64427378967024	9%
0.64427378967024	0.65244952181274	10%
0.65244952181274	0.66184657879897	11%
0.66184657879897	0.67125814175790	12%
0.67125814175790	0.68047193193158	13%
0.68047193193158	0.68830555161621	14%
0.68830555161621	0.69666020072924	15%
0.69666020072924	0.70388414439727	16%
0.70388414439727	0.71153298410291	17%
0.71153298410291	0.71899454804265	18%
0.71899454804265	0.72842289189164	19%
0.72842289189164	0.73652249238699	20%
0.73652249238699	0.74543636742703	21%
0.74543636742703	0.75351564051001	22%
0.75351564051001	0.76215264341252	23%
0.76215264341252	0.77078297978634	24%
0.77078297978634	0.77778005227287	25%
0.77778005227287	0.78611868125808	26%
0.78611868125808	0.79452033073732	27%
0.79452033073732	0.80227687182006	28%
0.80227687182006	0.81068840950830	29%
0.81068840950830	0.81940279406454	30%
0.81940279406454	0.82863334797177	31%
0.82863334797177	0.83699138271781	32%
0.83699138271781	0.84490126552511	33%

0.84490126552511	0.85263689336860	34%
0.85263689336860	0.86220561525508	35%
0.86220561525508	0.87170736326378	36%
0.87170736326378	0.88042590645531	37%
0.88042590645531	0.88934192133808	38%
0.88934192133808	0.89803565732055	39%
0.89803565732055	0.90760546566724	40%
0.90760546566724	0.91711399269845	41%
0.91711399269845	0.92665541420424	42%
0.92665541420424	0.93482914035115	43%
0.93482914035115	0.94504238184167	44%
0.94504238184167	0.95320582391708	45%
0.95320582391708	0.96243427199884	46%
0.96243427199884	0.97141500347254	47%
0.97141500347254	0.97960925718515	48%
0.97960925718515	0.99082682863388	49%
0.99082682863388	1.000000000000001	50%
1.000000000000001	1.01065485901958	51%
1.01065485901958	1.02157391917041	52%
1.02157391917041	1.03181408023206	53%
1.03181408023206	1.04290709369499	54%
1.04290709369499	1.05300343580120	55%
1.05300343580120	1.06334382623829	56%
1.06334382623829	1.07439584185308	57%
1.07439584185308	1.08577986242574	58%
1.08577986242574	1.09544918746511	59%

1.09544918746511	1.10705628384487	60%
1.10705628384487	1.11997738822194	61%
1.11997738822194	1.12942030426358	62%
1.12942030426358	1.14185541457440	63%
1.14185541457440	1.15234442314801	64%
1.15234442314801	1.16399551477489	65%
1.16399551477489	1.17656325837651	66%
1.17656325837651	1.19018468911221	67%
1.19018468911221	1.20213422269647	68%
1.20213422269647	1.21567283358572	69%
1.21567283358572	1.22791608907198	70%
1.22791608907198	1.24141082631924	71%
1.24141082631924	1.25377938288031	72%
1.25377938288031	1.26877072165772	73%
1.26877072165772	1.28193361666114	74%
1.28193361666114	1.29853083455681	75%
1.29853083455681	1.31014588815420	76%
1.31014588815420	1.32455174430608	77%
1.32455174430608	1.34049975511985	78%
1.34049975511985	1.35731773146750	79%
1.35731773146750	1.37463535009464	80%
1.37463535009464	1.39224716050131	81%
1.39224716050131	1.40773678250452	82%
1.40773678250452	1.42548384439408	83%
1.42548384439408	1.44443536742566	84%
1.44443536742566	1.46479968014262	85%

1.46479968014262	1.48742806263682	86%
1.48742806263682	1.50968118989544	87%
1.50968118989544	1.53317154635861	88%
1.53317154635861	1.56311479808836	89%
1.56311479808836	1.59606064434504	90%
1.59606064434504	1.62623982811626	91%
1.62623982811626	1.66260973272395	92%
1.66260973272395	1.69451827350295	93%
1.69451827350295	1.73922305956362	94%
1.73922305956362	1.79071556331901	95%
1.79071556331901	1.85341242865515	96%
1.85341242865515	1.93023256569179	97%
1.93023256569179	2.03513854165712	98%
2.03513854165712	2.20440690637111	99%
2.20440690637111	3.02787815706513	100%
3.02787815706513	99999999.0000000000000000	100%